



THROTTLE LK

Sri Lanka's Motorbike Marketplace

A Dedicated Motorcycle, Scooter, Spare Parts & Modified Parts Marketplace for Sri Lanka

BUSINESS & PROJECT PROPOSAL

Prepared by
PT Dilshan

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Prepared by	PT Dilshan
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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CDN	Content Delivery Network
JWT	JSON Web Token
MVP	Minimum Viable Product
RBAC	Role-Based Access Control
SEO	Search Engine Optimization
UI/UX	User Interface / User Experience
WAF	Web Application Firewall
LKR	Sri Lankan Rupee

1. Executive Summary

ThrottleLK is a proposed commercial digital marketplace dedicated to motorcycles, scooters, spare parts, modified parts, bike dealers and parts dealers in Sri Lanka. The business idea addresses a gap created by the fragmented way motorcycle buyers and sellers currently use general classifieds, social-media groups, individual dealer pages and physical shops.

The platform will provide a structured motorcycle-first experience rather than treating bikes as a generic advertisement category. Buyers will be able to search and filter by brand, model, variant, engine capacity, year, mileage, condition, price and location; compare listings; save favourites; communicate with sellers; and discover compatible parts and approved dealers. Private sellers, bike dealers and parts dealers will receive dedicated listing, inventory and showroom capabilities.

ThrottleLK will initially operate through a responsive, SEO-first Next.js web marketplace. A Flutter mobile application is planned as a major product channel and will reuse the same NestJS REST API and PostgreSQL data platform. This API-first design allows the business to expand from web to mobile without rebuilding the core business logic.

The launch strategy is intentionally staged. The first objective is to establish a trustworthy marketplace with quality inventory, strong search, moderation, dealer onboarding and organic search visibility. Revenue can later be generated through promoted listings, dealer packages, premium showrooms, homepage placements, brand advertising and related lead-generation products.

2. Introduction

Motorcycles and scooters are an important part of personal and commercial mobility in Sri Lanka. The buying journey, however, is distributed across general classified websites, social-media communities, dealer yards and word-of-mouth networks. A buyer searching for a specific motorcycle often has to compare inconsistent advertisements manually, while sellers and dealers have limited tools to present structured inventory or measure buyer interest.

ThrottleLK is designed as a focused digital marketplace for this ecosystem. Its purpose is to improve discovery, trust and lead connection for motorcycles and related parts while creating a scalable digital business that can operate across web and mobile channels.

The platform does not initially attempt to complete the vehicle transaction or payment inside the system. Instead, it helps the buyer discover a suitable listing, understand its details, compare alternatives and contact the relevant seller or dealer through phone, WhatsApp or in-platform messaging.

3. Background and Market Opportunity

3.1 Motorcycle marketplace context

Motorcycle purchasing decisions depend on information that is more specific than a typical general-classified advertisement. Brand, model, variant, engine capacity, manufacture year, registration year, mileage, condition, colour, service history, modifications, accident disclosure, seller type and location can all influence a buyer's decision.

3.2 Current buying and selling environment

General classifieds provide broad reach, while social platforms provide informal community reach. Dealer websites and physical yards provide business-specific inventory. The opportunity for ThrottleLK is to bring these behaviours into a single motorcycle-focused discovery experience with consistent data, search and dealer presentation.

3.3 Existing approaches and opportunity gap

Approach	Strength	Limitation for this market	ThrottleLK response
General classifieds	Large general audience	Generic filters and mixed categories	Motorcycle-specific taxonomy and filters
Social-media groups / marketplace	Fast informal reach	Weak structured search, inconsistent listings	Structured inventory, moderation and permanent pages
Individual dealer websites	Strong dealer identity	Inventory is fragmented across sites	Dealer directory and unified showrooms
Physical yards / word-of-mouth	Local trust	Limited national discovery	Location-aware digital discovery and contact

Table 3.1 - Existing approaches and opportunity gap

3.4 Spare and modified parts opportunity

The motorcycle ecosystem extends beyond complete vehicles. Spare parts and modified parts are often difficult to discover by compatibility. ThrottleLK therefore includes hierarchical part categories, bike-to-part fitment and parts-dealer showrooms so a user can identify which part suits a particular brand or model.

3.5 Business opportunity

A focused marketplace can build a defensible brand around motorcycle discovery, dealer connectivity and parts fitment. Permanent brand, model, category and location pages can also create an organic acquisition channel through search engines, reducing long-term dependence on paid traffic.

4. Problem Statement

The central problem is the absence of a single motorcycle-first digital marketplace in Sri Lanka that combines structured bike listings, advanced motorcycle search, dealer inventory, parts compatibility, moderation, bilingual access and long-term search visibility.

- Buyers spend time moving between multiple sources and manually comparing inconsistent listings.
- Private sellers have limited guidance for creating complete motorcycle-specific advertisements and measuring interest.
- Bike dealers lack a shared marketplace where a professional showroom and inventory can be discovered alongside private listings.
- Parts buyers struggle to identify compatible spare or modified parts and suitable parts dealers.
- Marketplace quality can be reduced by duplicate, misleading, expired or suspicious listings without a clear moderation workflow.
- Generic listing pages do not fully exploit permanent brand/model/location pages for organic search acquisition.

5. Vision, Aim and Objectives

5.1 Vision

To build a leading Sri Lankan digital ecosystem for motorcycle buying, selling, parts discovery and dealer connectivity across web and mobile platforms.

5.2 Aim

The aim of ThrottleLK is to design, launch and grow a fast, trustworthy and motorcycle-specific marketplace that connects buyers, private sellers, bike dealers and parts dealers while maintaining a scalable technical foundation for future commercial expansion.

5.3 Specific objectives

- Provide structured motorcycle and scooter browsing, search and filtering using motorcycle-specific attributes.
- Enable private sellers to create, edit, pause, renew and mark listings as sold under a moderated publishing workflow.
- Provide approved bike dealers with public showrooms, inventory management, leads and performance visibility.
- Create a dedicated spare and modified parts marketplace with categories, fitment and parts-dealer profiles.
- Provide buyers with favourites, comparison, saved searches, notifications and messaging.
- Provide administrators with control over listings, users, dealers, taxonomy, reports, promotions and advertising.
- Provide English and Sinhala public experiences.
- Build permanent, indexable brand, model, category, location and listing pages for organic search growth.
- Use one versioned REST API for both the Next.js web application and the planned Flutter mobile application.
- Launch on cost-conscious infrastructure and scale individual components only when usage justifies it.

5.4 Key deliverables

- Responsive public web marketplace
- NestJS REST API and PostgreSQL database
- Seller and dealer dashboards
- Spare/modified parts marketplace
- Administration and moderation console
- SEO landing-page structure
- Flutter mobile application path using the same API
- Deployment, monitoring and operational documentation

6. Stakeholder Analysis

Stakeholder	Primary need	Main concern	Design response
Guest / registered buyer	Find suitable bikes and parts quickly	Scattered listings and weak comparison	Structured search, filters, compare, favourites and contact tools
Private seller	Reach serious buyers	Low visibility and incomplete ads	Guided listing flow, images, status management and performance data
Bike dealer	Manage and promote inventory	Fragmented online presence	Approved showroom, inventory, leads and analytics
Parts dealer	Promote compatible parts and shop	Poor digital discovery	Parts inventory, fitment, profile and map presence

Stakeholder	Primary need	Main concern	Design response
Administrator	Maintain quality and trust	Spam, fraud and inconsistent taxonomy	Moderation, reports, approvals and platform controls
Advertiser / partner	Reach relevant motorcycle audience	Unfocused advertising	Marketplace placements and future targeted commercial products

Table 6.1 - Stakeholder analysis

7. Proposed Solution - The ThrottleLK System

7.1 Product overview

ThrottleLK will operate as a discovery and lead-connection marketplace. Users can browse public inventory, search by motorcycle-specific criteria and contact sellers. Transactions are completed outside the platform during the initial business phase.

7.2 User roles

- Guest buyer
- Registered buyer
- Private seller
- Approved bike dealer
- Approved parts dealer
- Administrator

7.3 Main buyer journey

Google/direct visit -> home, brand, model or category page -> search and filter -> listing details -> favourite/compare/share/message -> seller or dealer profile -> WhatsApp/call/in-app message -> deal completed outside the platform.

7.4 Seller journey

Register and verify -> create listing -> upload photographs -> preview -> submit for review -> administrator approval -> publish -> receive inquiries -> pause, renew or mark as sold.

7.5 Listing lifecycle

Draft -> Pending Review -> Active or Rejected. An active listing can later become Paused, Sold or Expired. Sold pages should remain available with a sold state and relevant alternatives rather than being immediately deleted.

8. Core Marketplace Features

- Home page with marketplace search, categories, featured inventory and promotions.
- Bike browse pages and category pages for scooters, street/commuter, high-capacity/sports, trail/adventure, classic/cruiser and electric motorcycles.
- Filtering by brand, model, variant, category, price, manufacture/registration year, mileage, engine capacity, fuel type, transmission, location, seller type and condition.
- Sorting by newest, oldest, price, mileage, year and popularity.
- Detailed listing pages with price, negotiable status, service history, modifications, accident declaration, location and multiple images.
- Side-by-side motorcycle comparison.

- Buyer favourites and saved searches.
- Notifications and in-platform conversations.
- Seller profiles, listing management and listing-performance information.
- Phone and WhatsApp contact controls.

9. Spare & Modified Parts Marketplace

ThrottleLK will include a dedicated marketplace for spare parts and modified parts rather than mixing these items into vehicle listings. Parts will use hierarchical categories and fitment relationships so users can understand which brand/model a part is intended to suit.

- Part listings and images
- Spare-part and modified-part categories
- Brand/model fitment
- Part favourites and inquiries
- Parts-dealer profiles
- Parts-dealer map directory
- Parts inventory and performance information

10. Dealer Ecosystem

Business sellers require capabilities beyond a single private listing. ThrottleLK will therefore provide an approval-based dealer model. Dealers apply for a business profile, complete their information, pass administrator review and then manage a public showroom and inventory.

- Dealer application and approval
- Public showroom
- Multiple inventory listings
- Dealer contact and lead capture
- Listing and inventory performance
- Map/location discovery
- Promotion requests and future dealer packages

11. Administration & Marketplace Control

- Marketplace overview and analytics
- Bike and part listing moderation
- User management
- Bike-dealer and parts-dealer approval
- Brand, model, category, district and city taxonomy
- User reports and suspicious-listing review
- Promotion packages and requests
- Homepage advertising placements
- Platform configuration and health monitoring

12. Business Model & Monetization

The early marketplace can remain largely free to reduce friction in acquiring inventory and buyers. Monetization should be introduced only after the platform demonstrates useful traffic, dealer participation and lead generation.

- Featured, top and urgent listing products

- Dealer monthly packages
- Premium dealer and parts-dealer showrooms
- Homepage and category advertising
- Motorcycle brand and accessory advertising
- Upgraded dealer analytics
- Future finance and insurance lead placements where commercially and legally appropriate

Pricing is intentionally not fixed in this proposal. Package design and price points should be validated with real dealers and parts businesses before commercial launch.

13. SEO & Digital Growth Strategy

SEO is a core acquisition strategy for the web platform. Important pages should be server-rendered or statically generated rather than relying only on a client-side application. Permanent pages can target searches around motorcycle brands, models, categories and locations.

- Permanent brand and model landing pages
- Location-aware model pages
- Unique listing URLs with permanent identifiers
- Canonical URLs and selective noindex rules for faceted search
- Page-specific titles and meta descriptions
- Open Graph metadata
- Valid structured data where the visible content supports it
- XML sitemap and robots controls
- Buying guides and internal linking

Examples of target URL patterns include `/bikes`, `/bikes/honda`, `/bikes/honda/hornet-160r` and `/listing/<descriptive-slug>-<id>`.

14. System Architecture

The platform follows an API-first architecture. The Next.js web application and planned Flutter mobile application are separate clients that consume the same NestJS REST API. PostgreSQL remains the system of record; Redis is used only where caching or coordination provides a measurable benefit; media files are stored in object storage.

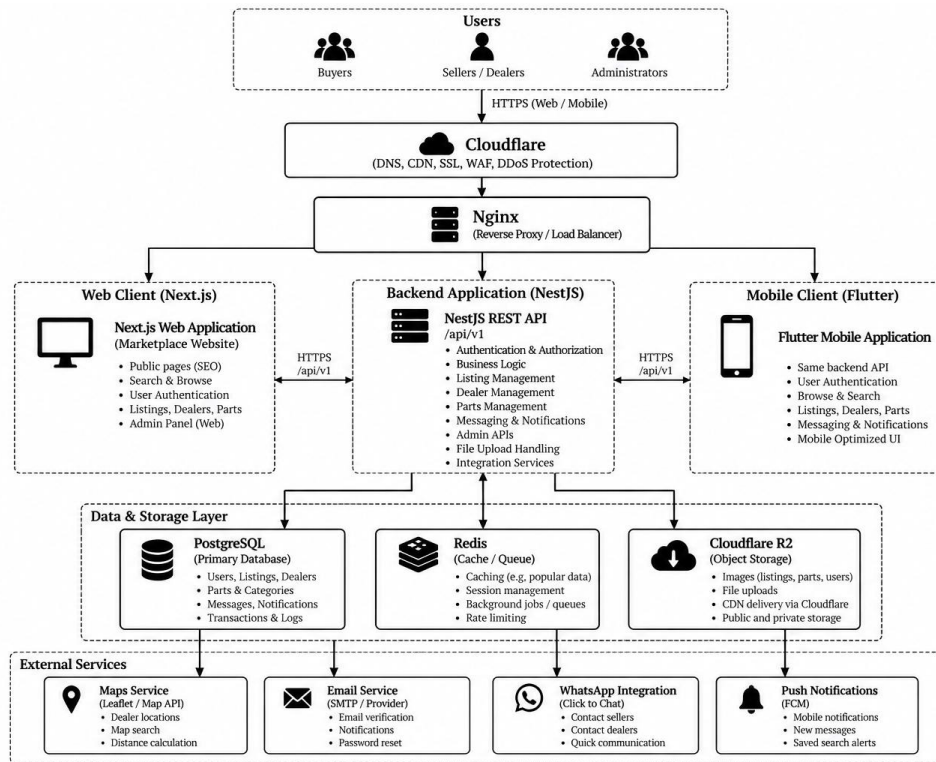


Figure 14.1 - High-level architecture of the ThrottleLK platform

For web traffic, Cloudflare can provide DNS, SSL, CDN and WAF capabilities in front of Nginx. Nginx routes requests to the relevant Node.js processes. The backend remains stateless where practical so additional API instances can be introduced later without redesigning the business logic.

15. Technology Stack

Layer	Technology	Purpose
Public web	Next.js 15, React, TypeScript	SEO-capable responsive marketplace
Mobile application	Flutter, Dart	Native-style Android/iOS client using the same REST API
Web styling	Tailwind CSS	Consistent responsive UI
Backend	NestJS	Modular business logic and REST API
ORM	TypeORM	PostgreSQL mapping and migrations
Database	PostgreSQL	Primary system of record
Cache / coordination	Redis	Caching, rate-limit support and later jobs
Authentication	JWT access + refresh sessions, Passport	Authentication and session control
Storage	S3-compatible object storage / Cloudflare R2	Listing and profile images
Maps	Leaflet on web; suitable Flutter map integration	Dealer and parts-dealer discovery
Security	Helmet, CORS, throttling, Cloudflare	Application and edge protection
Monitoring	Sentry / structured logs	Error tracking and operational visibility
Runtime / deployment	Node.js, Nginx, PM2	Application hosting and process management

Table 15.1 - Technology stack

16. Database & Data Architecture

PostgreSQL will store the operational marketplace data. Important entity groups include identity and access, marketplace taxonomy, bike listings, dealers, parts listings, communications, trust and promotions.

- Identity: users, roles, refresh sessions and authentication tokens.
- Taxonomy: brands, bike models, categories, districts and cities.
- Bike marketplace: listings, listing images, inquiries, engagement events, favourites and saved searches.
- Dealers: dealer profiles, images, inventory items and related documents.
- Parts marketplace: part categories, part listings, images, fitments, favourites, inquiries and parts dealers.
- Communication and growth: reports, notifications, conversations, messages, promotion packages, requests and homepage placements.

Indexes should follow real search patterns such as listing status, brand, model, district, price and published date. As inventory grows, cursor/keyset pagination is preferred over very large offset-based pagination.

17. Security, Privacy & Trust

- HTTPS-only production traffic
- Validated input for all mutating APIs
- Role-based authorization
- Hashed passwords and short-lived access tokens
- Refresh-token rotation
- Rate limiting on sensitive endpoints
- Strict image type and size validation
- Secrets stored outside source control
- Separate development, staging and production configuration
- Minimal public personal information
- Seller-controlled phone and WhatsApp visibility
- No public NIC, full private home address, password hashes, refresh tokens or internal moderation notes

Trust is also a product function. Listing review, dealer approval, reporting, expiry handling and visible listing states help the marketplace maintain quality as inventory grows.

18. Web & Mobile Strategy

The web platform is the first acquisition channel because it supports search-engine indexing, shareable public pages and low-friction access without installation. The Flutter application is planned as the dedicated mobile channel for repeat users and can add richer mobile experiences such as push notifications in later releases.

Both channels will reuse the same versioned NestJS API, authentication model and PostgreSQL data. This prevents separate business logic from developing independently in web and mobile applications and reduces future maintenance cost.

19. Scope and Development Priorities

Priority	Functions	Reason
Core / MVP	Authentication; taxonomy; bike listings; images; search/filter; listing details; seller management; moderation; responsive web; core SEO	Establishes the complete discovery-to-contact marketplace flow
Secondary	Favourites; compare; saved searches; messaging; notifications; dealer showrooms; parts marketplace; maps; analytics	Improves retention, supply-side value and ecosystem depth
Mobile expansion	Flutter buyer/seller experience using the same API	Adds a dedicated mobile channel after core APIs and journeys are stable
Future	Paid checkout, finance/insurance integrations, advanced recommendations, AI listing assistance, fraud/image intelligence, dedicated search engine	Deferred until business demand and platform scale justify the complexity

Table 19.1 - Scope prioritization

20. Implementation Roadmap

Development follows an iterative, business-first sequence. The core marketplace should be usable before secondary features and monetization layers are expanded.

- Foundation: repository, environments, CI, database and design system.
- Core API: authentication, users, taxonomy, listings, media and moderation.
- Public web: home, browse, search, listing details, seller flow and responsive UI.
- SEO: permanent brand/model/location pages, metadata, sitemap and structured data.
- Buyer retention: favourites, comparison, saved searches, notifications and messaging.
- Dealer and parts ecosystem: dealer applications, showrooms, inventory, parts, fitment and maps.
- Flutter mobile application: reuse stable APIs and business rules.
- Growth and hardening: promotions, analytics, caching, security, performance and operational monitoring.

21. Testing & Quality Assurance

Quality assurance will cover both functional correctness and marketplace experience.

Area	Evaluation
Unit testing	Services, validation, taxonomy helpers and listing-status rules
Integration testing	Authentication, listing creation, moderation, favourites, dealer and parts workflows
API smoke testing	Critical routes verified after deployment
End-to-end testing	Register -> create listing -> approve -> search -> contact -> mark sold
SEO QA	Titles, canonicals, sitemap, robots, structured data and status codes
Performance	Home, search, model and listing pages under increasing load
Security	Authentication, authorization, uploads, rate limits and security headers

Area	Evaluation
Accessibility	Keyboard access, labels, focus, contrast and image alt text
Usability	Task-based review with representative buyers, sellers and dealers

22. Operational & Technical Feasibility

Dimension	Assessment	Risk / mitigation
Technical	The selected web, API, database and Flutter technologies are mature and suitable for staged implementation.	Use modular architecture and MVP-first delivery.
Operational	Marketplace workflows can begin with manual dealer approval and listing moderation.	Automate only after operational volume justifies it.
Economic	The platform can launch on modest infrastructure and scale components separately.	Avoid over-provisioning and measure bottlenecks before spending.
Market	The concept addresses fragmented motorcycle, dealer and parts discovery.	Validate demand through pilot dealers, real listings and buyer usage.
Schedule	Core web and API work can precede mobile and advanced growth features.	Protect the core scope and defer nonessential features.
Privacy / trust	The system can minimize public personal data and control contact visibility.	Apply moderation, reporting, RBAC and responsible data handling.

Table 22.1 - Feasibility summary

23. Business Risks & Mitigation

Risk	Potential impact	Mitigation
Low listing supply at launch	Buyers see little value	Onboard initial dealers and seed genuine quality inventory
Low buyer traffic	Sellers receive few leads	SEO landing pages, content, targeted launch marketing and partnerships
Fake or stolen listings	Loss of trust	Moderation, reports, image rules, expiry and dealer approval
Spam / scraping	Operational and infrastructure abuse	Rate limits, Cloudflare WAF and CAPTCHA when justified
Scope creep	Delayed launch	Maintain Core / Secondary / Future boundaries
Premature infrastructure cost	Poor unit economics	Scale only measured bottlenecks
Weak parts fitment data	Incorrect discovery	Admin-controlled taxonomy and verified fitment workflows
Mobile/web feature divergence	Higher maintenance cost	One API and shared business contracts

Table 23.1 - Risk assessment

24. Expected Business Outcomes

24.1 Buyers

- Faster motorcycle and parts discovery
- Better comparison using motorcycle-specific information

- Clearer distinction between private sellers and dealers
- Saved searches, favourites and communication tools
- English and Sinhala access

24.2 Sellers and dealers

- Structured listings aligned with buyer search behaviour
- Professional digital showrooms
- Inventory and lead visibility
- Promotion opportunities
- A dedicated motorcycle audience

24.3 ThrottleLK business

- A focused motorcycle marketplace brand
- Organic traffic through permanent SEO pages
- Reusable backend for web and Flutter mobile channels
- Multiple future monetization paths
- Scalable architecture without high launch infrastructure cost

25. Future Expansion

- Expanded Flutter application with push notifications
- Paid dealer subscriptions and promoted-listing checkout
- Deeper spare-parts and accessories ecosystem
- Finance and insurance enquiry partnerships
- Dealer team accounts
- Improved recommendation engine
- AI-assisted listing descriptions
- Duplicate-image and fraud-risk detection
- Dedicated search engine such as OpenSearch or Typesense if PostgreSQL search becomes insufficient

26. Conclusion

ThrottleLK is proposed as a commercial, motorcycle-first digital marketplace for Sri Lanka. Its value is not simply the ability to post an advertisement; it is the combination of structured motorcycle data, focused search, dealer and parts ecosystems, moderation, permanent search-friendly pages and a technical foundation that can serve both web and mobile users.

The initial business objective is to launch a fast and trustworthy marketplace that can acquire quality listings, buyers and dealers. The platform can then expand into promotions, dealer subscriptions, advertising, richer mobile experiences and related services without rebuilding its core API and data architecture.

By keeping the launch scope disciplined and measuring real marketplace behaviour before scaling infrastructure or advanced features, ThrottleLK can develop from a focused web marketplace into a broader motorcycle digital ecosystem.

Appendix A - Plan of Action

The following ten-month plan provides a practical staged path from business validation and requirements through marketplace development, Flutter mobile implementation, quality assurance and pilot launch. The schedule can be adjusted based on validation results and available development capacity.

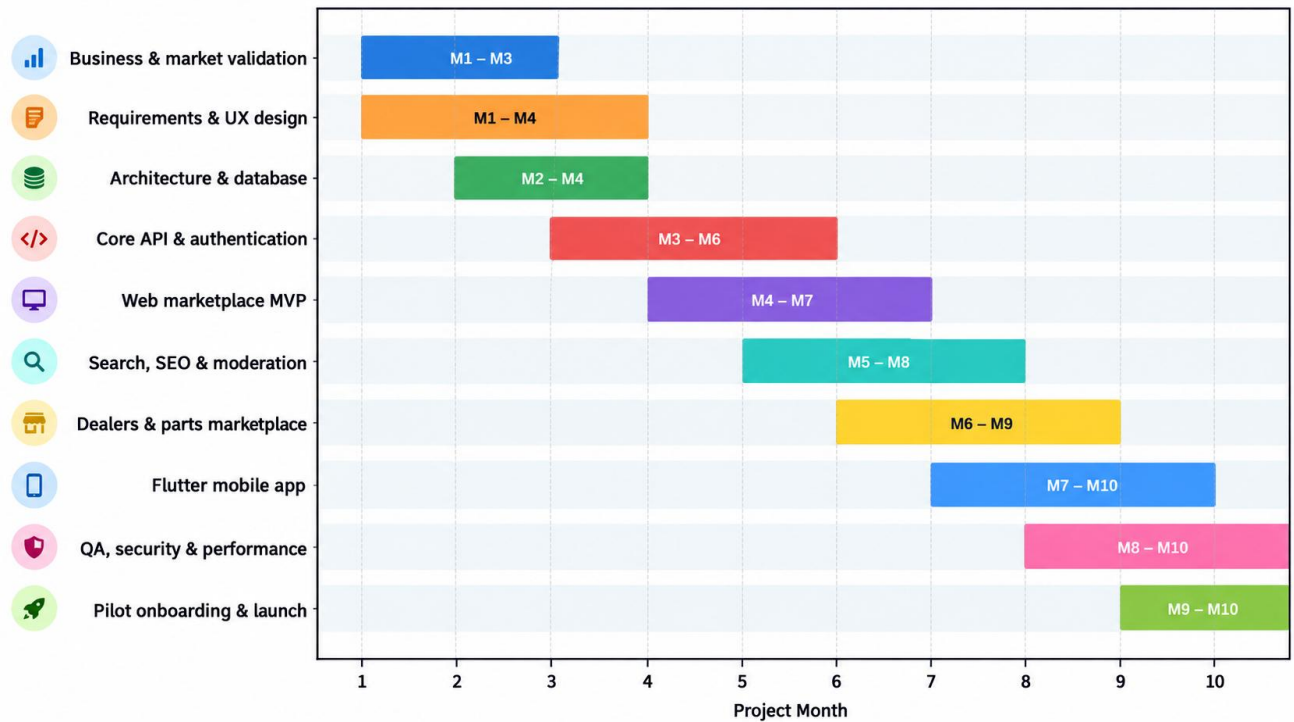


Figure A.1 - Ten-month implementation plan

Appendix B - Initial Motorcycle Categories

Initial categories: scooter; street / commuter; high-capacity / sports; trail / adventure; classic / cruiser; electric motorcycle and scooter. Additional categories remain administrator-configurable.

Appendix C - Core API Groups

```

/api/v1/auth/*
/api/v1/users/me
/api/v1/listings
/api/v1/brands
/api/v1/models
/api/v1/locations
/api/v1/dealers
/api/v1/favourites
/api/v1/saved-searches
/api/v1/reports
/api/v1/part-listings
/api/v1/parts-dealers
/api/v1/conversations
/api/v1/notifications
/api/v1/admin/*

```